BIG DATA ANALYTICS PROJECT

NYC Traffic & Mobility Analytics

An end-to-end pipeline for NYC taxi trips

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INTRODUCTION

Agenda

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Problem, dataset, zones

02 · Architecture
Design and tech stack

03 · Infrastructure
Docker and HDFS storage

04 · Spark ETL
Ingestion, cleaning, outputs

05 · Orchestration
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06 · Analytics
ClickHouse and Superset

07 · Machine Learning
XGBoost models, predictor

08 · Streaming Pipeline
Kafka and Spark Streaming

INTRODUCTION

Motivation and Goals

- **Millions** of NYC taxi trips are recorded every month
- Raw data is large and inconsistent, so it needs cleaning first
- We turn it into insights: trips, revenue, speed and duration
- We reveal the busiest zones and hours of the day
- We predict trip duration, fare and zone demand



INTRODUCTION

The Dataset

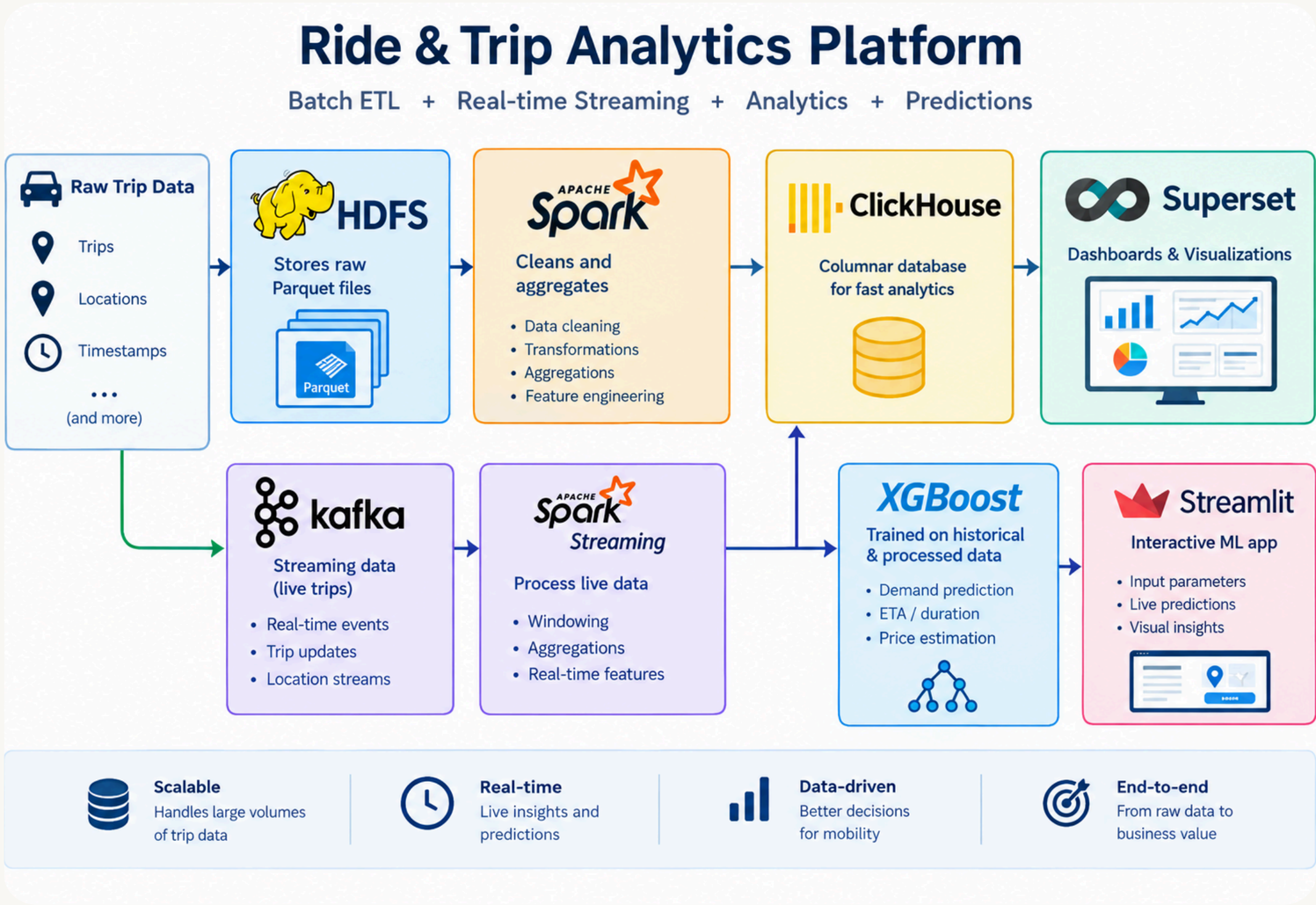
- **NYC yellow-taxi trips**, one Parquet file per month
- Key fields: times, passengers, distance, zones, payment, fare, tip
- Schemas differ between months, so each file is aligned before merging
- **Taxi Zone Lookup**: 265 zones mapped to borough and zone name

Column Name	Description
VendorID	TPEP provider (1=CMT,
tpep_pickup_datetime	Pickup time
tpep_dropoff_datetime	Dropoff time
passenger_count	Passenger count
trip_distance	Trip distance
RatecodeID	Final rate code (1-6)
store_and_fwd_flag	Store and forward flag (N/Y)
PULocationID	Pickup zone ID
DOLocationID	Dropoff zone ID
payment_type	Payment type (1-6)
fare_amount	Time-and-distance fare
extra	Extras and surcharges
mta_tax	MTA tax ($0.50)
tip_amount	Tip amount
tolls_amount	Total tolls
improvement_surcharge	Improvement surcharge ($0.30)
total_amount	Total amount charged
congestion_surcharge	Congestion surcharge
airport_fee	Airport fee

ARCHITECTURE

System Architecture

- **HDFS** stores raw Parquet files
- **Spark** cleans and aggregates
- **Airflow** orchestrates the ETL
- **ClickHouse + Superset**: dashboard
- **XGBoost + Streamlit**: predictions
- **Kafka + Spark Streaming**: live trips



ARCHITECTURE

Technology Stack

Layer	Tool	Role
Storage	HDFS (Hadoop 3.3.6)	Raw and processed data
Processing	Apache Spark 3.5	ETL, cleaning, aggregation
Orchestration	Airflow 2.10 + PostgreSQL 15	Pipeline DAG and run state
Analytics DB	ClickHouse 24.8	Fast aggregate queries
Dashboards	Apache Superset 6.0	KPIs and charts
Machine learning	XGBoost	Duration, fare, zone-hour models
Application	Streamlit	Interactive trip predictor
Streaming	Kafka 4.3 + Spark Streaming	Live trip events
Packaging	Docker Compose	One-command environment

ARCHITECTURE

HDFS Storage

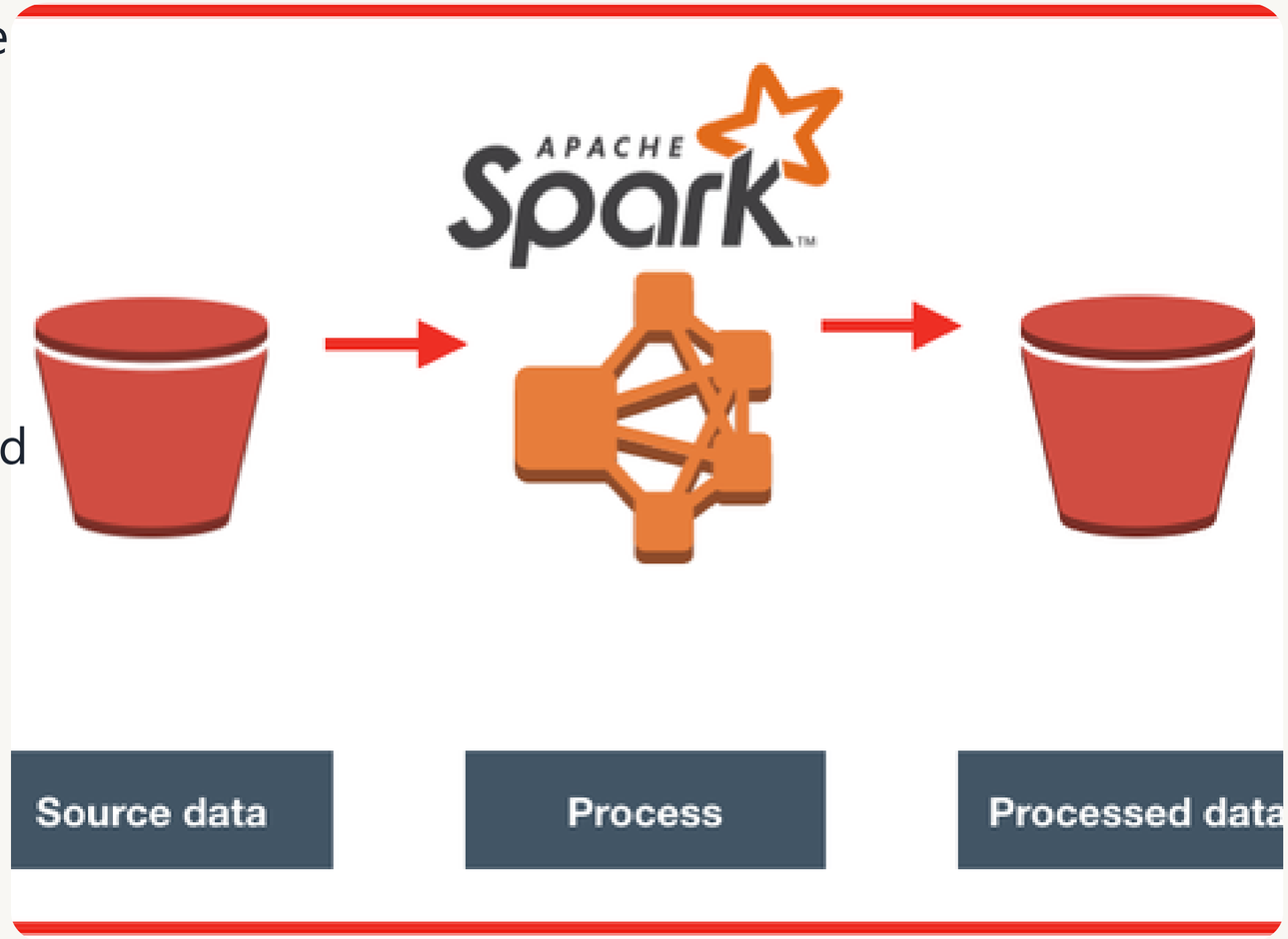
- **NameNode + DataNode** with replication factor 1
- **/data/nyc/raw**: original monthly Parquet files
- **/data/nyc/processed**: cleaned trips and zone-hour data
- **/data/nyc/staging**: one isolated folder per Airflow run
- Raw data is never modified, so every output can be rebuilt



DATA PIPELINE

Spark ETL: Ingestion and Features

- Lists every raw Parquet file and reads each one separately
- Aligns column types, then merges files by column name
- Renames columns to a consistent, readable schema
- Adds duration, date, hour, weekday, month and year
- Adds the average speed of each trip



DATA PIPELINE

Data Quality and Cleaning

- A **quality report** counts violations per rule, and overlaps, before removal
- Missing zones or timestamps; pickup year outside 2021–2026
- Duration outside 1–180 min; distance outside 0–100 miles
- Fare or total not positive; passengers outside 1–6
- Average speed above 80 mph

DATA PIPELINE

ETL Outputs

- **trips**: cleaned trips, partitioned by year and month
- **zone_hour**: one row per pickup zone, date and hour
- Metrics: trips, fare, distance, duration, speed, revenue
- Summary: raw, clean and removed rows, plus removal rate
- Top 5 busiest zone-hour groups are printed

DATA PIPELINE

Airflow Orchestration

- **One DAG**, manual trigger, no retries, 3-hour ETL limit
- Checks HDFS and Spark are up, then runs preflight
- Preflight: output folder must be new and raw data readable
- Each run writes to its own staging folder
- Validation: trips and zone-hour outputs exist and have rows



ANALYTICS

ClickHouse Analytics Layer

- Zone-hour aggregates are loaded into **ClickHouse**
- Columnar database built for fast aggregate queries
- Columns: zone, date, hour, trips, fare, distance, duration, speed, revenue
- Zone lookup turns IDs into readable zone names
- Superset connects through a ClickHouse driver



ANALYTICS

Superset Dashboard

- **KPIs**: total trips, revenue, average speed and duration
- **Trends**: daily trips and trips by hour
- **Rankings**: top 10 pickup zones by trips and by revenue
- **Filters**: time range, pickup hour and pickup zone
- Answers where and when demand and revenue peak



MACHINE LEARNING

Machine Learning Overview

- **XGBoost** gradient-boosted trees: 300 trees, depth 8
- Three questions: how long, how much, how busy
- Train before Oct 2025, test from Oct 2025 onward
- Random samples keep training light on one machine
- Evaluated with MAE, RMSE and R^2, saved with each model



MACHINE LEARNING

Trip Duration and Fare Models

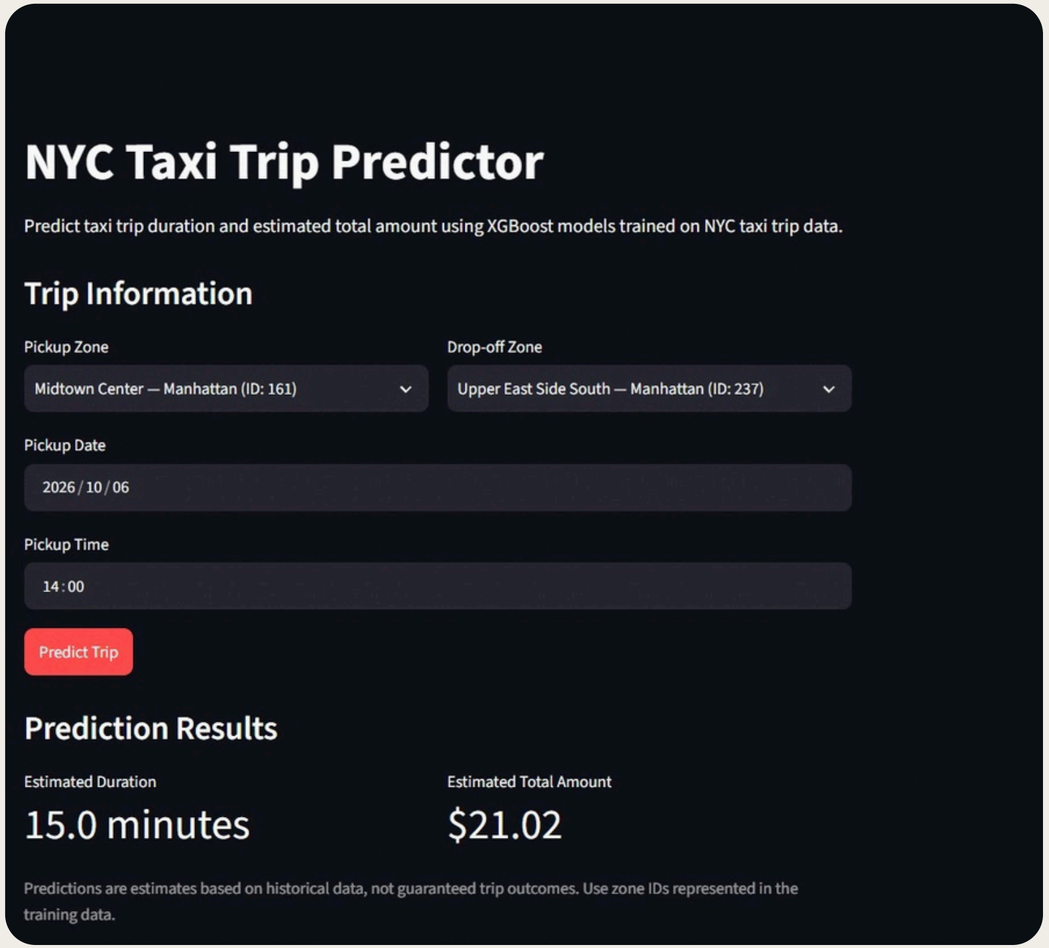
Model	Features	Duration	Fare
With distance	Zones, passengers, distance, time	R^2 0.79 · MAE 4.2 min	R^2 0.90 · MAE $3.8
Without distance	Zones, time	R^2 0.56 · MAE 6.4 min	R^2 0.76 · MAE $6.8

- Time features: hour, weekday, month, year, weekend
- About 1.3M training and 277K test trips

MACHINE LEARNING

Streamlit Trip Predictor

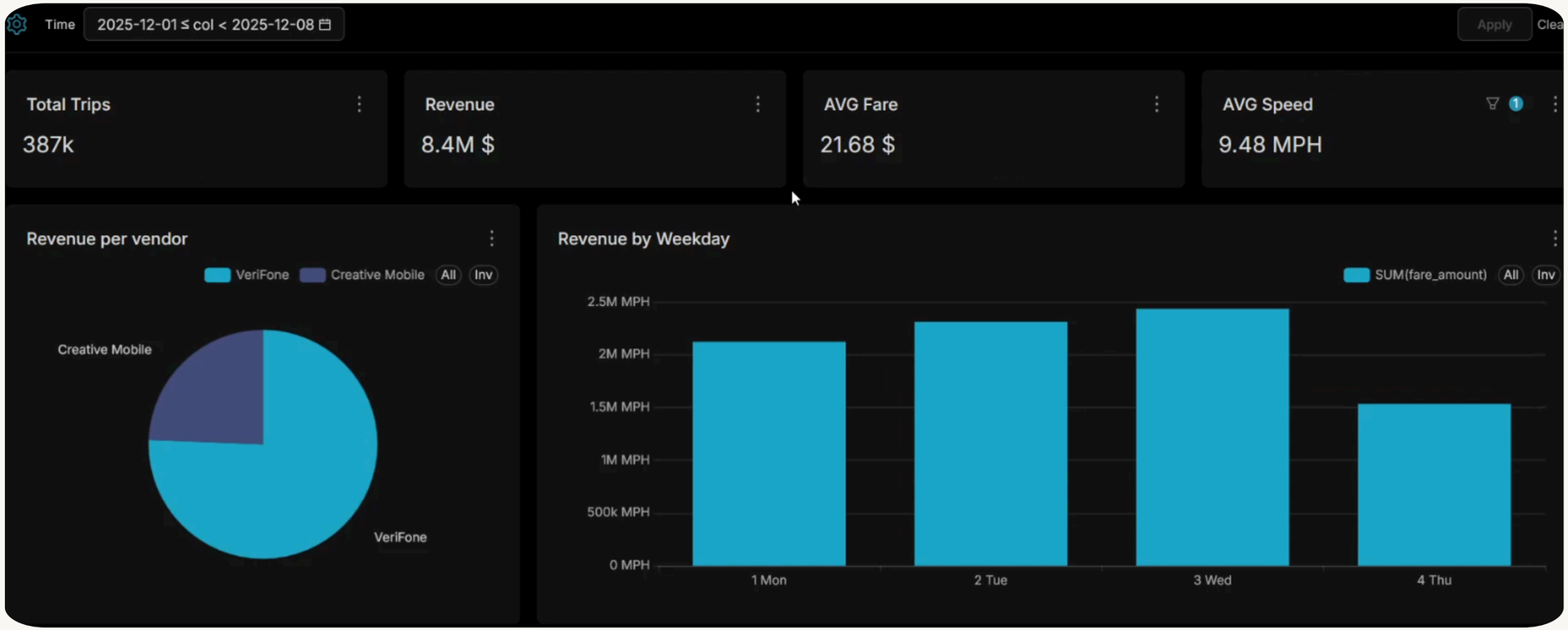
- **Inputs**: pickup and drop-off zone, distance, passengers, date, time
- **Outputs**: estimated duration and total amount
- Uses the saved XGBoost models and the zone lookup
- Shows trip summary, model inputs and an estimate disclaimer
- Runs locally, no cluster needed



REAL-TIME STREAMING

Real-Time Streaming Overview

- **One table**, **trips_live**, serves the whole dashboard
- Holds raw trip fields, derived numbers and text labels
- **Superset 6.1** reads it directly; labels become chart dimensions
- New rows arrive with every batch while the stream runs



Thank you

Questions?

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